

Genetic Algorithm and Perceptron Model – Step-by-Step Solutions

Assessment Tool- Co-2 At- 1

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ITA0612- Machine Learning

1. Genetic Algorithm

Given:

Function: $f(x) = x^2$

Initial population: 2, 4, 6, 8

We need to:

1. Calculate fitness values
2. Select the best candidates
3. Perform crossover
4. Explain mutation

Step 1: Calculate Fitness Values

The fitness function is: $f(x) = x^2$

Individual (x)	Fitness $f(x) = x^2$
2	$2^2 = 4$
4	$4^2 = 16$
6	$6^2 = 36$
8	$8^2 = 64$

Therefore: $f(2) = 4$, $f(4) = 16$, $f(6) = 36$, $f(8) = 64$

Step 2: Select the Best Candidates

Since the objective is to maximize $f(x)$, we select individuals having the highest fitness.

$x = 8 \rightarrow \text{Fitness} = 64$

$x = 6 \rightarrow \text{Fitness} = 36$

Therefore, the selected parents are 8 and 6.

Step 3: Convert Parents into Binary

Using 4-bit binary representation:

$$8 = (1000)_2$$

$$6 = (0110)_2$$

Parent 1: 1000

Parent 2: 0110

Step 4: Perform Crossover

The question asks to exchange the least significant bits (LSBs). The LSB is the rightmost bit.

Parent 1: 1000 $\rightarrow$ LSB = 0

Parent 2: 0110 $\rightarrow$ LSB = 0

After exchanging the LSBs:

Offspring 1: 1000 = 8

Offspring 2: 0110 = 6

Therefore, Offspring = 8, 6.

Note: In this case, crossover does not change the offspring because both parents have the same LSB, which is 0.

Step 5: Role of Mutation

Mutation means randomly changing one or more bits in an offspring.

Example: 1000 $\rightarrow$ 1001

Here, the last bit changes from 0 to 1.

$$1001_2 = 9$$

$$\text{New fitness: } f(9) = 9^2 = 81$$

This is better than the previous maximum fitness of 64.

Importance of Mutation:

- Maintains diversity in the population.
- Introduces new solutions.
- Prevents the algorithm from getting stuck at a local optimum.

- Increases the possibility of finding a better/global optimum solution.

Final Answer – Genetic Algorithm

Fitness values = 4, 16, 36, 64

Best candidates = 8 and 6

Offspring after LSB crossover = 8 and 6

Mutation helps introduce new solutions and improves population diversity.

2. Perceptron Model

Given:

Advertisement exposure: $x_1 = 2$

Income level: $x_2 = 1$

Weight $w_1 = 0.5$

Weight $w_2 = 0.3$

Bias $b = -0.4$

Step 1: Perceptron Net Input Formula

The net input is:

$$z = w_1x_1 + w_2x_2 + b$$

Substitute the values:

$$z = (0.5)(2) + (0.3)(1) - 0.4$$

Step 2: Calculate

$$z = 1 + 0.3 - 0.4$$

$$z = 0.9$$

Therefore, the net input is 0.9.

Step 3: Apply Step Activation Function

Step activation function:

$$y = 1, \text{ if } z \geq 0$$

$$y = 0, \text{ if } z < 0$$

Since $z = 0.9 \geq 0$, therefore $y = 1$.

Step 4: Classification

1 = Customer purchases the product

0 = Customer does not purchase the product

Since the output is 1, the customer will purchase the product.

Final Answer – Perceptron

Net Input = 0.9

Output = 1

Therefore, the customer will purchase the product.